Sarvesh Suhas Kute

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EDUCATION

Vellore Institute of Technology, Vellore 2024 – 2028
Bachelor of Technology in Computer Science and Engineering **CGPA: 9.36**
Relevant Coursework: Data Structures and Algorithms, Design and Analysis of Algorithms, Database Systems, Computer Networks, Object-Oriented Programming, Computer Architecture and Organization, Theory of Computation, Discrete Mathematics

TECHNICAL SKILLS

Languages: Python, C, C++, Java, JavaScript, SQL
Backend & Web: Flask, Node.js, Express.js, React.js, Next.js, REST APIs, JWT Authentication, RBAC, Socket Programming (TCP/UDP), DNS
Databases: MySQL, PostgreSQL, Stored Procedures, Triggers, Transactions, Query Optimization, Indexing, Schema Design
Systems: Linux, Linux Kernel, eBPF, BCC, POSIX APIs, Operating Systems, Computer Networks
Tools & DevOps: Git, GitHub, GitHub Actions (CI), Docker, Vercel, Code Review, Unit and Integration Testing

EXPERIENCE

Software Engineering Intern – Supertech Instrumentation Services (I) Pvt. Ltd. May 2026 – Jul 2026
Thane, India (On-site and Remote)

- Built 12 REST API endpoints with Node.js, Express and PostgreSQL to serve batch history and sensor data to a React dashboard used by plant operators and engineers.
- Cut average API response time from 850 ms to 120 ms on tables with 500,000+ sensor readings by profiling queries with EXPLAIN ANALYZE and adding composite indexes on (batch_id, timestamp). Replaced unbounded alarm-log queries with keyset pagination.
- Designed 3 normalized tables and wrapped multi-table batch inserts in transactions, so batch and sensor data stay consistent when an insert fails.
- Implemented JWT authentication with role-based access (admin, engineer, operator), and wrote Jest and Supertest tests covering auth, role permissions and validation edge cases.
- Fixed a timezone bug that skewed daily energy totals by migrating to TIMESTAMPTZ (UTC storage, IST display). Replaced a manual Excel compliance process with server-side PDF reports.

PROJECTS

RouteWise | github.com/sarveshsuhaskute/RouteWise Flask, MySQL, Docker, GitHub Actions

- Designed a 19-table MySQL schema (BCNF) with 30 triggers, 8 stored procedures, 7 views and 20 CHECK constraints; seat booking, cancellation and transfer run as atomic transactions with row-level locking to prevent double-booking.
- Benchmarked indexing on 200,000 rows: 57×–257× faster queries, including FULLTEXT search vs LIKE (41.2 ms to 0.16 ms).
- Implemented server-enforced role-based access (viewer, staff, admin), a least-privilege database user, salted password hashing, CSRF protection, login lockout and HMAC-signed QR e-tickets.
- Built an end-to-end test suite and a GitHub Actions CI pipeline (lint, tests on real MySQL, Docker build, accessibility audit).

Network Service Platform | github.com/sarveshsuhaskute/network-service-platform Python, Sockets, DNS, Concurrency

- Built a concurrent server hosting four services: TCP echo, SHA-256-verified file transfer, UDP echo, and an RFC 1035 DNS server (A, AAAA, CNAME, MX, TXT) with upstream forwarding and caching, compatible with dig.
- Added idle timeouts, client limits, simulated packet loss, rotating logs, and a web dashboard for load testing; wrote 33 functional, fault-injection and security tests plus a benchmark harness.

Linux File Prefetcher using eBPF | github.com/sarveshsuhaskute/ai-file-prefetcher- Python, BCC, eBPF, C, Linux Kernel

- Built a context-aware prefetch daemon that traces file access with eBPF tracepoints and a ring buffer, learns access sequences with an online Markov model, and preloads files using posix_fadvise.
- Evaluated against default kernel readahead across four baselines: matched its 99.25% hit rate under normal load, and traced a 26% throughput loss under memory pressure to userspace overhead and eviction blindness.
- Diagnosed a ctypes 64-bit pointer truncation bug that silently unmapped the wrong memory region.